A Boundary Layer Model for 1-DTE Options

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Abstract. We introduce a boundary-layer-motivated stochastic volatility model for one-day-to-expiration (1-DTE) options. The model consists of a baseline Heston-type variance process together with a jump-activated fast mean-reverting factor that represents short-lived intraday departures from ordinary volatility dynamics. We also develop an empirical procedure for identifying these fast-regime periods from intraday option-surface data using local changes in at-the-money implied volatility, skew, and curvature. Simulations show that the model reproduces transient volatility bursts and heavy-tailed minutely returns. In SPY 1-DTE data, the proposed detector identifies concentrated intervals of rapid surface deformation consistent with the two-scale structure. We then study the effect of this regime information in a delta-hedging experiment. Although the resulting strategy does not substantially improve upon a fully rehedged Black–Scholes benchmark in raw hedge error, it does improve the tradeoff between hedge accuracy and hedge frequency by concentrating rehedging effort in fast periods. The results support the use of short-scale corrections in the analysis of ultra-short-maturity options.

Key words. Boundary Layer Theory, Matched Asymptotics, Stochastic Volatility, 1-DTE, Options

1. Introduction.

One-day-to-expiration (1-DTE) options lie in an extreme short-maturity regime in which prices and hedge ratios can change rapidly over the course of a single trading session. Near expiry, small movements in the underlying and small local deformations of the implied-volatility surface can produce large changes in option sensitivities[3]. For this reason, ultra-short-maturity options are not always well described by modeling approaches that treat intraday volatility dynamics as evolving on a single homogeneous scale.

A recurring empirical feature of intraday 1-DTE data is that large movements in at-the-money implied volatility are often accompanied by abrupt changes in local skew and curvature. These episodes are typically brief, concentrated, and economically important for pricing and hedging. A baseline Heston-type model captures stochastic variance and leverage effects, but it does not explicitly separate ordinary intraday variation from short-lived bursts in which the option surface deforms much more rapidly[6]. This motivates a two-scale formulation with a slow background component and a transient fast component[4].

In this paper, we introduce a boundary-layer-motivated stochastic volatility model for 1-DTE options. The model augments a Heston-type variance process with a jump-activated fast mean-reverting factor designed to represent short-lived intraday departures from ordinary volatility dynamics. To connect the model with data, we also construct an empirical fast-regime detector based on unusually large local changes in at-the-money implied volatility, skew, and curvature in intraday SPY 1-DTE options. These labels serve as observable proxies for periods in which the short-scale component is most relevant.

We study the framework from three directions. First, we examine the qualitative behavior of the model in simulation and show that it can generate transient volatility bursts and

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heavy-tailed minutely returns. Second, we apply the detection procedure to SPY 1-DTE data and identify concentrated intervals of rapid local surface deformation. Third, we test whether the resulting regime information is useful for dynamic hedging. On the tested sample, a fully rehedged Black–Scholes benchmark attains the smallest raw one-minute hedge error. However, the regime-aware strategy yields nearly the same hedge accuracy as the fully rehedged smile-based strategy while using substantially fewer hedge updates. Thus, in the present implementation, the main value of the fast-regime signal is in the timing of hedge adjustments rather than in a large improvement in the hedge ratio itself.

The contribution of the paper is therefore twofold. On the modeling side, it proposes a tractable two-scale stochastic-volatility specification tailored to ultra-short-maturity options. On the empirical side, it develops a practical procedure for identifying transient fast periods directly from intraday option-surface data and shows that this signal is useful for hedge scheduling.

The remainder of the paper is organized as follows. Section 2 reviews the analytical and financial motivation for the model. Section 3 presents the model formulation. Section 4 studies its qualitative behavior in simulation. Section 5 develops the data pipeline and fast-regime detector. Section 6 reports the hedging experiment. Section 7 and section 8 contain the discussion and conclusions.

2. Motivation and Background.

The proposed model is motivated by the interaction of two features of ultra-short-maturity options. The first is analytical. As expiry approaches, option prices and sensitivities become increasingly dependent on local behavior near the strike and near the terminal time. In such settings, short-time asymptotics and boundary-layer ideas are natural tools for describing local corrections to a leading-order approximation [7, 2]. Our paper does not derive a full matched asymptotic expansion. Rather, these ideas motivate a two-scale stochastic-volatility specification in which a baseline slow component is supplemented by a short-lived fast correction.

The second feature is empirical. Implied-volatility surfaces do not move by rigid parallel shifts; instead, they evolve through changes in local level, slope, and curvature. This is especially relevant for 1-DTE options, where small intraday deformations of the smile can materially change option values and hedge ratios. In the SPY data studied later in the paper, the minutes identified as stressed periods are associated with abrupt joint movements in at-the-money implied volatility, skew, and butterfly curvature; see section 5. This suggests that a single-scale volatility diffusion may miss the transient local deformations that are most important for near-expiry contracts.

A natural starting point is therefore a Heston-type stochastic-volatility model, which captures stochastic variance and leverage effects [6]. However, a baseline Heston model remains a single-scale description. To represent short-lived departures from ordinary intraday behavior, it is natural to introduce an additional fast component, in the spirit of multiscale stochastic-volatility models [8]. Recent work on 1-DTE options also supports the view that the very short end of the maturity spectrum should be studied with methods tailored to local short-horizon structure rather than as a minor perturbation of standard medium-maturity models.

These considerations motivate the model introduced in section 3. We retain a slow Heston-type variance process as the background dynamics and augment it with a jump-activated fast

mean-reverting factor. The role of this factor is not to replace the baseline stochastic-volatility structure but to supply a transient short-scale correction during those intervals in which the local smile evolves much more rapidly than usual. The empirical regime labels developed in section 5 are then used as observable proxies for the activation of this fast component.

3. Model Formulation.

We introduce the following modeling:

$$\frac{\mathrm{d}S_t}{S_t} = \mu \mathrm{d}t + \sqrt{\nu_t}\mathrm{d}W_t^S$$

$$\mathrm{d}\nu_t = (\kappa(\theta - \nu_t) + \frac{A}{\varepsilon}Y_t)\mathrm{d}t + \xi\sqrt{\nu_t}\mathrm{d}W_t^\nu + \xi\sqrt{\nu_t}\frac{B}{\sqrt{\varepsilon}}Y_t\mathrm{d}W_t^J$$

$$\mathrm{d}Y_t = -\frac{\gamma}{\varepsilon}Y_t\mathrm{d}t + \mathrm{d}J_t$$

$$\begin{bmatrix}\mathrm{d}W_t^S\\ \mathrm{d}W_t^\nu\\ \mathrm{d}W_t^J\end{bmatrix} \sim \mathcal{N}(\vec{0}, \mathbf{\Sigma}) \qquad \mathbf{\Sigma} = \mathrm{d}t\begin{bmatrix}1 & \rho & \rho\\ \rho & 1 & \rho_J\\ \rho & \rho_J & 1\end{bmatrix}$$

where (W_t^S, W_t^ν, W_t^J) is three-dimensional Brownian motion with a correlation matrix described above, and J_t is a Poisson process with parameter $\lambda \in \mathbb{R}^+$. Further, $\rho, \rho_J \in [-1, 1]$, $A, \mu, \in \mathbb{R}$, and $\xi, \theta, \kappa, B, \nu_t, \gamma, \varepsilon \in \mathbb{R}^+$ with $\varepsilon \ll 1$.

The model builds on the Heston model, assuming that there are distinct regimes of behavior of the underlying. Unlike the traditional Heston model, which assumes a relatively small distribution of mean-reverting volatility, this model allows for a regime shift over a time scale $O(\varepsilon)$ triggered by the Poisson process J_t. The process Y_t serves to incorporate these "fast"—high volatility and high volatility of volatility—characteristics into volatility. Further, in order for the "fast" regime to return to the "slow" regime, we have the influence of the Poisson process decay exponentially over time, ensuring the traditional behavior of the Heston model over time scales $O(1)$.

For our simulations, we utilize a $\mathrm{d}t = 1$ minute. Further, a common error with simulating the CIR model is that ν_t may become negative due to the discretization of our time step. To account for this, if ν_t is ever negative, we resample ν_t until it is nonnegative.

4. Model Expressivity.

We provide key characteristics that this model produces that are distinct to 1-DTE options. To demonstrate the model's expressivity, we utilize the ad-hoc parameters outlined in Table 2, simulating at a 1-minute time scale.

As illustrated in Figure 1, the jump process intermittently triggers a"fast regime," represented by the state variable Y_t spiking to 1.0 before exponentially decaying. These fast-regime entries directly translate into abrupt spikes in realized annualized volatility, which quickly revert to the slow-regime baseline. Consequently, the underlying asset S_t experiences wide, immediate price swings. This transient, highly volatile behavior closely mirrors the intraday dynamics traditionally observed in 1-DTE options, particularly near the market open, close, and around significant scheduled announcements.

The inclusion of these fast mean-reverting shocks significantly effects the kurtosis of the asset's returns. Figure 2 shows the distribution of volatility and log-returns across 10^5 simulations, highlighting how this regime-switching mechanism successfully generates heavy-tailed

distributions for volatility and log-returns. Displaying an excess kurtosis of ≈ 3.7 and a massive Jarque-Bera statistic ($> 2.2 \times 10^7$), ($p << 10^{-15}$), the model realistically captures the leptokurtic distribution of one minute returns that standard SDE models do not normally reproduce.

Beyond time-series properties, the boundary layer model produces the correct implied volatility smile. As shown in Figure 3, the regime-switching volatility induces a pronounced volatility skew across strike prices. Furthermore, the simulation accurately captures market microstructure effects; applying a standard \$0.01 minimum bid to the generated prices results in an artificial inflation of implied volatility at extreme out-of-the-money (OTM) strikes.

Finally, the model offers significant structural advantages for risk management as expiration approaches. Figure 4 compares the boundary layer model to a standard Black-Scholes benchmark (using the slow regime's mean volatility of 12.1%) at exactly 30 minutes to expiry. While the shape of the option payoffs look similar, the boundary layer model has well-defined Greeks. Because the boundary layer model explicitly accounts for volatility blow-ups and heavier tails in the fast regime, it produces well-defined deltas and gammas near expiry. Conversely, the traditional Black-Scholes gamma at-the-money incorrectly approaches infinity. This framework therefore provides a more robust model to price and dynamically hedge 1-DTE options

5. Data and Fast-Regime Identification.

Our empirical fast-regime labels are constructed from an intraday panel of SPY 1-DTE option quotes derived from the workbook `Option_Data.xlsx`. The preprocessing, implied-volatility inversion, feature construction, and regime labeling follow the deterministic pipeline implemented in `IV.py`, `Butterfly.py`, and `Regime_Label.py`. The pipeline first, stored in `pipeline.ipynb`, builds a cleaned option quote panel with implied volatilities, then computes minute-level smile diagnostics, and finally converts large local surface moves into binary fast-regime labels [1].

5.1. Intraday quote panel and 1-DTE filtering.

The workbook is parsed sheet-by-sheet into a long quote panel. Each contract is stored in five columns,

$$\text{Dates} \mid \text{Open} \mid \text{Close} \mid \text{Value} \mid \text{Volume},$$

with the first row containing the contract name and the second row containing field labels. Contract metadata, including expiry, call/put flag, and strike, are extracted directly from the contract name. Because the workbook does not contain bid and ask quotes, the close price is used as the transaction-price proxy and, for all downstream calculations, as the option mid. An intraday spot series is then inferred from matched call–put pairs using put–call parity. For a common timestamp bin, expiry, and strike,

$$C(t) - P(t) = S(t) - Ke^{-r\tau},$$

so that

$$S(t) = C(t) - P(t) + Ke^{-r\tau}.$$

In the implementation we take $r = 0$, which yields the working approximation

$$S(t) \approx C(t) - P(t) + K.$$

To obtain stable intraday matches, quote times are first rounded to 10-second bins, calls and puts with the same (t, expiry, K) are paired, and the resulting parity-implied spot values are aggregated using a robust volume-weighted average.

After spot inference, the option panel is synchronized to a regular one-minute grid. For each contract, quotes are carried backward to the minute timestamp with an as-of merge subject to a quote-age restriction, so that the final panel is dense but does not rely on excessively stale prices. Time to expiry is computed both in minutes and in year-fraction form, and we retain only rows for which the trade date is exactly one calendar day before expiry. Thus the panel used in this section is a pure 1-DTE panel.

5.2. Implied volatility and local smile coordinates. For each synchronized quote, implied volatility is obtained by numerically inverting the Black–Scholes price map for the observed option mid. The same step also yields Black–Scholes vega, which is later used as a reliability weight in local cross-sectional aggregation. If K denotes strike and $S(t)$ the parity-implied spot, we define log-moneyness by

$$k = \log\left(\frac{K}{S(t)}\right).$$

The output of this stage is therefore a contract-level panel indexed by timestamp and containing, at minimum, $(k, \text{IV}, \text{vega})$ for each 1-DTE quote.

The fast-regime detector does not fit a full parametric surface at each minute. Instead, it uses local diagnostics built from the implied-volatility cross section near the money. This is deliberate as for 1-DTE options, the local shape of the smile can shift sharply over a few minutes, so a low-dimensional local summary is more robust than a global fit for event detection. The feature construction in `Butterfly.py` follows a three-point, wing-based approximation centered at the money with fixed wing location

$$H = 0.01$$

in log-moneyness and neighborhood tolerance

$$\text{TOL} = 0.0025.$$

At each minute, quotes are first collapsed to one implied volatility per $(t, k\text{-bin})$ by vega-weighted averaging, after which the local smile is sampled at $k = -H$, $k = 0$, and $k = +H$. Denoting the resulting local levels by

$$\sigma_L(t) \approx \sigma(t, -H), \qquad \sigma_0(t) \approx \sigma(t, 0), \qquad \sigma_R(t) \approx \sigma(t, +H),$$

we define the intraday smile summaries by

$$\sigma_{\text{atm}}(t) := \sigma_0(t),$$

$$\text{skew}(t) := \frac{\sigma_R(t) - \sigma_L(t)}{2H},$$

and

$$\text{bfly}(t) := \frac{\sigma_L(t) + \sigma_R(t)}{2} - \sigma_0(t), \qquad \text{curvature}_{\text{bfly}}(t) := \frac{2\,\text{bfly}(t)}{H^2}.$$

The quantity σ_{atm} records the local volatility level, skew is the central-difference approximation to $\partial\sigma/\partial k$ at the money, and $\text{curvature}_{\text{bfly}}$ is the corresponding second-difference proxy. In the code these three series are computed minute-by-minute and stored as the core inputs to the event detector.

5.3. Event score construction. Fast-regime identification is based on unusually large local movements in σ_{atm}, skew, and $\text{curvature}_{\text{bfly}}$. Each trading day is treated as its own intraday experiment. Within a day, each feature is first smoothed by an exponentially weighted moving average with span

$$\text{EWM_SPAN} = 7,$$

and then differenced at the one-minute level. For a generic feature $X(t)$, this gives a local increment series $\mathrm{d}X(t)$. Each increment series is then standardized by a robus z-score computed from the within-day median and median absolute deviation,

$$z_X(t) = \frac{dX(t) - \text{median}(dX)}{1.4826\,\text{MAD}(dX)},$$

with a standard deviation fallback if the MAD degenerates numerically. This normalization prevents a few large intraday outliers from dominating the scale of the entire day.

Abrupt persistent changes are then identified with a two-sided CUSUM detector. If $z(t)$ is one of the robustly standardized series, the detector updates positive and negative cumulative sums according to

$$C_t^+ = \max\{0,\, C_{t-1}^+ + z(t) - \kappa\}, \qquad C_t^- = \min\{0,\, C_{t-1}^- + z(t) + \kappa\},$$

with reference value

$$\kappa = 0.5$$

and threshold

$$h = 6.0.$$

A feature-specific event is recorded whenever $C_t^+ > h$ or $C_t^- < -h$, at which point the CUSUM statistics are reset. This yields three feature-specific event indicators,

$$\text{event}_{\sigma_{\text{atm}}}(t), \qquad \text{event}_{\text{skew}}(t), \qquad \text{event}_{\text{curvature}}(t).$$

The aggregate event indicator is then

$$\text{event_any}(t) = \max\{\text{event}_{\sigma_{\text{atm}}}(t), \text{event}_{\text{skew}}(t), \text{event}_{\text{curvature}}(t)\}\,,$$

and the aggregate intensity score is

$$\text{event_score}(t) = |z_{\sigma_{\text{atm}}}(t)| + |z_{\text{skew}}(t)| + |z_{\text{curvature}}(t)|.$$

Thus the event score is a direct summary of the size of the intraday standardized moves across the three local smile features.

5.4. Hysteresis labeling of fast regimes. The final step maps event intensity into a binary regime label

$$\text{regime}(t) \in \{0, 1\},$$

where 0 denotes the slow regime and 1 the fast regime. This is done by a hysteresis rule implemented in `Regime_Label.py`. A day begins in the slow regime. The process enters the fast regime when either

$$\text{event_score}(t) \geq 8$$

or

$$\text{event_any}(t) = 1.$$

Once the fast regime has been entered, it is held for at least

$$\text{HOLD_BARS} = 10$$

one-minute bars. During this hold period, or whenever the score remains above the low threshold,

$$\text{event_score}(t) \geq 4,$$

the regime remains fast. Exit occurs only after the score has stayed below the low threshold for long enough to exhaust the hold counter. This produces contiguous fast-regime windows and suppresses isolated one-minute label oscillations.

This labeling rule is closely aligned with the modeling intuition of section 2 and section 4. The slow regime represents the baseline intraday evolution of the local smile, while the fast regime marks short-lived intervals in which the ATM level, slope, or curvature moves in a way that is statistically exceptional relative to the rest of that day. In this sense, the empirical fast label serves as a minute-level proxy for the short-scale perturbation in the boundary-layer model.

5.5. Example: 2026–01–28. Figure 5 shows the extracted feature series for 2026–01–28. The top panel plots the ATM implied-volatility proxy $\sigma_{\text{atm}}(t)$, the next two panels plot the local skew and local butterfly curvature, and the bottom panel plots the aggregate event score. The shaded regions are the minutes classified as fast regime by the hysteresis rule. The example illustrates the main empirical pattern motivating our model. Fast-regime windows seem to align with localized bursts in the implied-volatility level together with abrupt deformations of slope and curvature. The detector therefore picks up rapid local changes in the shape of the 1-DTE smile itself. The same regime labels are then used downstream in the calibration and hedging pipeline.

6. Hedging Experiment.

In this section we estimate the parameters of the boundary layer model from the regime-labeled intraday panel and then test whether the fast-regime signal is useful for dynamic hedging. The main empirical question is not only whether the model reduces hedge error, but whether it improves the tradeoff between hedge accuracy and the number of hedge updates.

6.1. Parameter Estimation. We begin with the regime-labeled intraday panel constructed in section 5. The fast-regime intervals are identified from the event score built from ATM implied volatility, skew, and butterfly curvature. We then split the sample into slow-regime and fast-regime observations.

For the slow component, we remove all minutes classified as fast regime and use the remaining observations to estimate the baseline parameters μ, κ, θ, ξ. Let

$$v_t := \sigma_{\mathrm{atm}}(t)^2$$

denote the local variance proxy derived from the ATM implied volatility. On the slow-regime sample, we fit the baseline Heston-type variance dynamics

$$\mathrm{d}\nu_t = \kappa(\theta - \nu_t)\mathrm{d}t + \xi\sqrt{\nu_t}\mathrm{d}W_t^{\nu}$$

in discrete time using one-minute transitions.

For the fast component, we isolate the observations inside fast-regime windows and estimate the transient parameters $A, B, \varepsilon, \gamma, \lambda$. These parameters govern the short-scale perturbation

$$\mathrm{d}\nu_t = \left(\kappa(\theta - \nu_t) + \frac{A}{\varepsilon}Y_t\right)\mathrm{d}t + \xi\sqrt{\nu_t}\mathrm{d}W_t^{\nu} + \xi\sqrt{\nu_t}\frac{B}{\sqrt{\varepsilon}}Y_t\mathrm{d}W_t^{J}, \qquad \mathrm{d}Y_t = -\frac{\gamma}{\varepsilon}Y_t\mathrm{d}t + \mathrm{d}J_t.$$

Here A controls the extra drift during fast periods, B controls the additional diffusion loading, γ controls the decay back to the slow regime, and λ is the arrival intensity of fast-regime episodes.

Using the rebuilt regime-labeled panel, the estimated slow-regime parameters are

$$\mu \approx 0.1441, \qquad \kappa \approx 367.2932, \qquad \theta \approx 0.029256, \qquad \xi \approx 1.6341.$$

For the fast component, we obtain

$$A \approx 7.1911 \times 10^{-6}, \qquad B \approx 0.05282, \qquad \varepsilon \approx 1.0175 \times 10^{-5},$$

$$\gamma \approx 0.04533, \qquad \lambda \approx 1456.0 \text{ per year.}$$

Equivalently, the jump intensity corresponds to about

$$\lambda \approx 6.0 \text{ fast-regime entries per trading day.}$$

The estimation sample contains 24 trading days and 9720 total minute observations, of which 6400 slow-regime transitions are used for the baseline fit and 3296 fast-regime transitions are used for the transient fit. In total we identify 144 fast-regime entries.

A normalization is required to identify ε and γ separately from the decay of fast episodes. In the present calibration we set

$$\varepsilon = \frac{1}{252 \cdot 390},$$

which corresponds to one trading minute in year units, and then estimate γ from the decay of excess variance within fast windows. Under this normalization, the effective decay rate is

$$\frac{\gamma}{\varepsilon},$$

which corresponds to an estimated minute-scale decay of roughly

$$0.04533 \text{ per trading minute.}$$

These parameter estimates are then used to guide the hedging experiment below.

6.2. Backtest Design. We next run a new stock-only hedging experiment directly on the rebuilt intraday panel. The sample is constructed from the previous trading session for expiries from 2026–01–29 through 2026–02–09. For each expiry, we infer the underlying spot from put–call parity, rebuild the local smile features $(\sigma_{\text{atm}}, \text{skew}, \text{curvature}_{\text{bfly}})$, reapply the fast-regime classifier, and select one liquid near-the-money contract per day with stable one-minute observations. After filtering, the backtest contains 6 trade dates and 2366 one-minute option transitions.

For an option value process V_t and stock hedge h_t, the one-step hedge error is defined by

$$e_{t+1} := (V_{t+1} - V_t) + h_t(S_{t+1} - S_t).$$

We summarize each method using the mean absolute hedge error

$$\text{MAE} := \frac{1}{N} \sum_{t=1}^{N} |e_t|,$$

together with the root mean squared error, the 95% quantile of absolute error, the number of rehedges, and stock turnover.

We compare seven rules.

1. `bs_1m`: Black–Scholes delta using the contract's own implied volatility, rehedged every minute.
2. `surface_1m`: delta computed from the local quadratic smile

$$\sigma(k) = \sigma_{\text{atm}} + \text{skew}\, k + \tfrac{1}{2}\, \text{curvature}_{\text{bfly}}\, k^2,$$

   rehedged every minute.
3. `surface_5m`: the same smile-based delta, rehedged every five minutes.
4. `surface_10m`: the same smile-based delta, rehedged every ten minutes.
5. `surface_regime`: the smile-based delta, rehedged every minute in the fast regime and every ten minutes in the slow regime.
6. `surface_regime_band`: the same as `surface_regime`, but with an additional stock-position band of 0.08.
7. `shock_surface_regime`: the regime-dependent schedule together with an ad hoc fast-regime volatility uplift tied to the event score.

The purpose of this design is to separate two effects. The first is the effect of changing the delta formula, for example replacing a Black–Scholes delta by a local smile-based delta. The second is the effect of changing the timing of rehedges, in particular by concentrating hedge updates inside fast-regime windows.

Table 1

Summary of the custom hedging backtest on the rebuilt intraday sample.

Method	MAE	RMSE	95% abs. err.	Rehedges	Stock turnover
`bs_1m`	0.03841	0.06193	0.13152	2366	17.34
`surface_1m`	0.03972	0.06300	0.13819	2366	24.22
`surface_regime`	0.03973	0.06291	0.13531	1152	15.38
`surface_regime_band`	0.03976	0.06297	0.13531	1153	15.37
`surface_5m`	0.04009	0.06346	0.13640	484	9.33
`shock_surface_regime`	0.04017	0.06390	0.13753	1152	16.65
`surface_10m`	0.04038	0.06397	0.13758	246	6.80

6.3. Backtest Results. Table 1 reports the overall results. The smallest raw mean absolute hedge error is obtained by `bs_1m`, with

$$\mathrm{MAE} = 0.03841.$$

Thus, on this sample, moving from a Black–Scholes delta to a local smile-based delta does not improve the fully rehedged benchmark.

The more interesting comparison is instead between the smile-based rules. Relative to `surface_1m`, the regime-aware rule `surface_regime` cuts the number of rehedges from 2366 to 1152, which is a reduction of about 51.3%, while the MAE changes only from

$$0.03972 \quad \text{to} \quad 0.03973.$$

The banded version behaves almost identically. This indicates that, for this sample, the main value of the fast-regime signal is in the timing of hedge updates rather than in large additional position-band effects.

The lower-frequency rules `surface_5m` and `surface_10m` compress trading much further, to 484 and 246 rehedges, respectively, but at the cost of somewhat larger hedge error. Relative to `surface_1m`, the MAE rises by about 0.9% for `surface_5m` and by about 1.7% for `surface_10m`. Hence the dominant frontier in this experiment is mainly a frontier between hedge intensity and rehedge frequency.

The shock-adjusted strategy `shock_surface_regime` does not improve the tradeoff. Its MAE is larger than that of `surface_regime` at the same rehedge count. On this sample, then, the fast-regime signal helps mainly by changing the hedge schedule rather than by materially changing the delta itself.

6.4. Interpretation. Figure 6 plots the efficiency frontier in the plane of MAE versus total rehedge count. The figure shows that the Black–Scholes one-minute hedge attains the smallest raw MAE, but it does so at the cost of rehedging at every opportunity. More importantly, the regime-aware smile hedge lies very close to the fully rehedged smile hedge while using roughly half as many updates. This is the main empirical gain from the fast-regime signal.

Figure 7 explains why. The regime-aware strategies hedge essentially every minute in the fast regime but only about 9% of slow-regime minutes. Their fast-regime hedge error is therefore nearly the same as that of `surface_1m`, while the trading savings come almost entirely from backing off in slow periods. In other words, the regime labels act as a useful timing device: they identify when aggressive intraday hedging is needed and when it is not.

Taken together, the results suggest the following. First, for this data set, Black–Scholes delta hedging remains a strong benchmark. Second, incorporating the local smile alone is not enough to reduce raw one-minute hedge error. Third, the fast-regime classifier is still useful because it improves the tradeoff between hedge error and trading activity. The main role of the boundary-layer structure in this backtest is therefore operational rather than purely computational: it tells the hedger when to hedge more aggressively.

7. Discussion. The results of this paper support the view that 1-DTE option markets are best described by at least two distinct intraday scales. On most minutes, the local smile evolves in a relatively regular manner and is reasonably represented by a slow mean-reverting volatility component. However, this baseline is repeatedly interrupted by short-lived episodes in which the ATM implied volatility, skew, and butterfly curvature move abruptly and jointly. These episodes are precisely the type of localized, fast variation that boundary layer arguments are meant to capture. In that sense, the empirical regime labels developed in section 5 are not merely a statistical device; they provide direct evidence that the observed 1-DTE surface contains a transient inner regime superimposed on a slower outer regime.

From a modeling standpoint, the simulations in section 4 show that the proposed process is able to reproduce several features that are difficult to obtain from simpler models. In particular, the model generates temporary bursts in realized volatility, heavy-tailed minutely returns, and a pronounced implied-volatility skew. These features are qualitatively consistent with the observed behavior of near-expiry options. The role of the fast factor is especially important near the money and near expiry, where small changes in local variance translate into large changes in option sensitivities. This gives a practical interpretation to the asymptotic motivation: the fast component acts as a short-scale correction that becomes relevant exactly where standard near-expiry approximations are most fragile.

The data analysis also suggests that fast-regime periods are not rare in an economically trivial sense. Although each individual episode is short, the detector identifies repeated intervals during the trading day in which local surface quantities move far more sharply than in ordinary periods. The fact that these labels are built jointly from level, slope, and curvature is important. Large moves in ATM implied volatility alone do not fully characterize the behavior of the 1-DTE surface. In practice, the smile often bends and twists at the same time that the ATM level moves, and these joint deformations are more closely aligned with the idea of a transient regime shift than any single feature taken in isolation. This is one reason the event score and hysteresis rule appear useful; they emphasize persistent local dislocations rather than isolated noisy observations.

The hedging experiment in section 6 gives a more nuanced conclusion. On the tested sample, a Black–Scholes delta hedge rebalanced every minute remains a strong benchmark in terms of raw hedge error. Thus the present boundary-layer implementation does not dominate a very aggressive baseline in the narrow sense of minimizing one-minute MAE. At the same time, the fast-regime signal still carries clear practical value. The regime-aware smile hedge achieves almost the same hedge accuracy as the fully rehedged smile hedge while using roughly half as many hedge updates. This suggests that the main empirical benefit of the framework is not necessarily a dramatic improvement in the instantaneous hedge ratio itself, but rather a better allocation of hedging effort across time. The fast-regime labels identify when intensive

rehedging is most needed and when slower updating is sufficient.

This point is important for interpretation. The present model should not be read as a claim that the market switches between two perfectly observable structural states. Rather, it provides a tractable approximation in which the slow regime describes ordinary intraday behavior and the fast regime marks periods when local surface dynamics are unusually unstable. In this framework, the empirical labels are best understood as proxies for the activation of the short-scale component Y_t. They need not coincide exactly with latent volatility shocks in the structural model to still be useful for calibration and hedging. Their value lies in linking the theoretical two-scale construction to something observable in the option surface itself.

There are several limitations to the present study. First, the data set is relatively small and is constructed from a workbook format that does not contain bid and ask quotes, so the close is used as the option-price proxy. This is a practical compromise and may introduce noise into both implied-volatility inversion and feature construction. Second, the regime-detection rule depends on fixed hyperparameters, such as the wing location H, the smoothing span, and the CUSUM thresholds. While these choices are reasonable and produce stable results on the sample studied here, alternative parameter choices could shift the exact timing and duration of fast-regime intervals. Third, the parameter estimation in subsection 6.1 is intentionally reduced-form. In particular, ε and γ are not separately identified without normalization, and the calibration should therefore be interpreted as a first-pass empirical fit rather than a definitive structural estimate.

A related limitation is that the hedging experiment tests only stock-only delta hedging over a short sample of near-the-money contracts. This is a natural first test because the main aim is to study whether the regime signal improves the timing of hedge updates. However, the model is richer than the present backtest fully exploits. Since the fast regime changes not only the variance level but also the local smile geometry, future work should examine whether the framework is more effective for hedging portfolios that are also sensitive to gamma, vega, or skew. It would also be useful to test the model on a longer sample and across different underlyings, especially in settings with known scheduled event risk.

There are also several natural mathematical extensions. One direction is to make the asymptotic connection more explicit by deriving a formal near-expiry expansion for option prices and Greeks under the proposed dynamics, rather than using the boundary-layer idea primarily as modeling guidance. Another is to replace the current empirical regime detector with a statistically estimated latent-state filter, allowing the fast component to be inferred jointly with the variance process. It may also be fruitful to allow the jump intensity λ or the fast-regime loading to vary systematically through the day, since the empirical results suggest concentration near open, close, and other high-information periods.

Overall, the main conclusion of the discussion is that the two-scale view is useful, even when the gains do not appear as a large reduction in headline hedge error. The evidence in this paper suggests that 1-DTE options are shaped by brief but important local surface shocks that are poorly summarized by a single-scale volatility model. The boundary-layer framework captures this qualitative structure, the regime detector identifies these episodes directly in the data, and the resulting signal improves the timing of dynamic hedging. For ultra-short-maturity options, that combination of theoretical structure, empirical identification, and operational usefulness is already a meaningful step.

8. Conclusions. We introduced a boundary-layer motivated stochastic volatility model for 1-DTE options that augments a baseline Heston-type variance process with a transient fast mean-reverting component. The model was designed to reflect the empirical fact that near-expiry option surfaces often alternate between ordinary intraday behavior and short-lived episodes of sharp local deformation in implied volatility, skew, and curvature.

The results of the paper support this two-scale view. In simulation, the model produces transient volatility bursts, heavy-tailed minutely returns, and a pronounced implied-volatility skew. In the data, the fast-regime identification procedure detects localized intervals in which the 1-DTE smile moves much more rapidly than in ordinary periods. In the hedging experiment, the resulting regime signal does not dramatically outperform a fully rehedged Black–Scholes benchmark in raw hedge error, but it does improve the tradeoff between hedge accuracy and hedge frequency by concentrating rehedging effort in fast periods.

Taken together, these findings suggest that a short-scale correction of the type proposed here is useful for studying and managing ultra-short-maturity options. The framework links asymptotic intuition, intraday option-surface data, and hedging practice in a single model. Future work should develop the asymptotic pricing formulas more explicitly, test the method on broader data sets, and study whether the fast-regime structure yields larger gains for richer hedging problems involving gamma, vega, or multi-option portfolios.

9. Acknowledgments. We thank Professor Welborn for his guidance, encouragement, and helpful feedback throughout the development of this project. Furthermore, we are deeply thankful for our fellow classmates peer review feedback. We also acknowledge the use of ChatGPT (GPT-5.4), Gemini (Pro) and Codex (5.3 Codex Spark) in the preparation of this paper. These tools were used to help identify coding bugs, assist with implementation and debugging, and improve wording and academic style during revision [10, 5, 9]. All modeling choices, analysis, and final presentation remain the responsibility of the authors.

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Appendix A. Model Expressivity Data & Graphs.

Table 2

Ad-hoc Parameters used to Demonstrate Model Expressivity. Unless otherwise specified, all parameters are in minute time scales because we performed our simulations with $\mathrm{d}t = 1$ *min.*

S_0	μ	ε	κ	θ	ξ	γ	A	B	ρ	ρ_J	λ
100	5% / year	3E−1	1E−2	1.5E−7	3E−5	1E−1	5E−10	2E−1	−0.7	0.6	2 / day

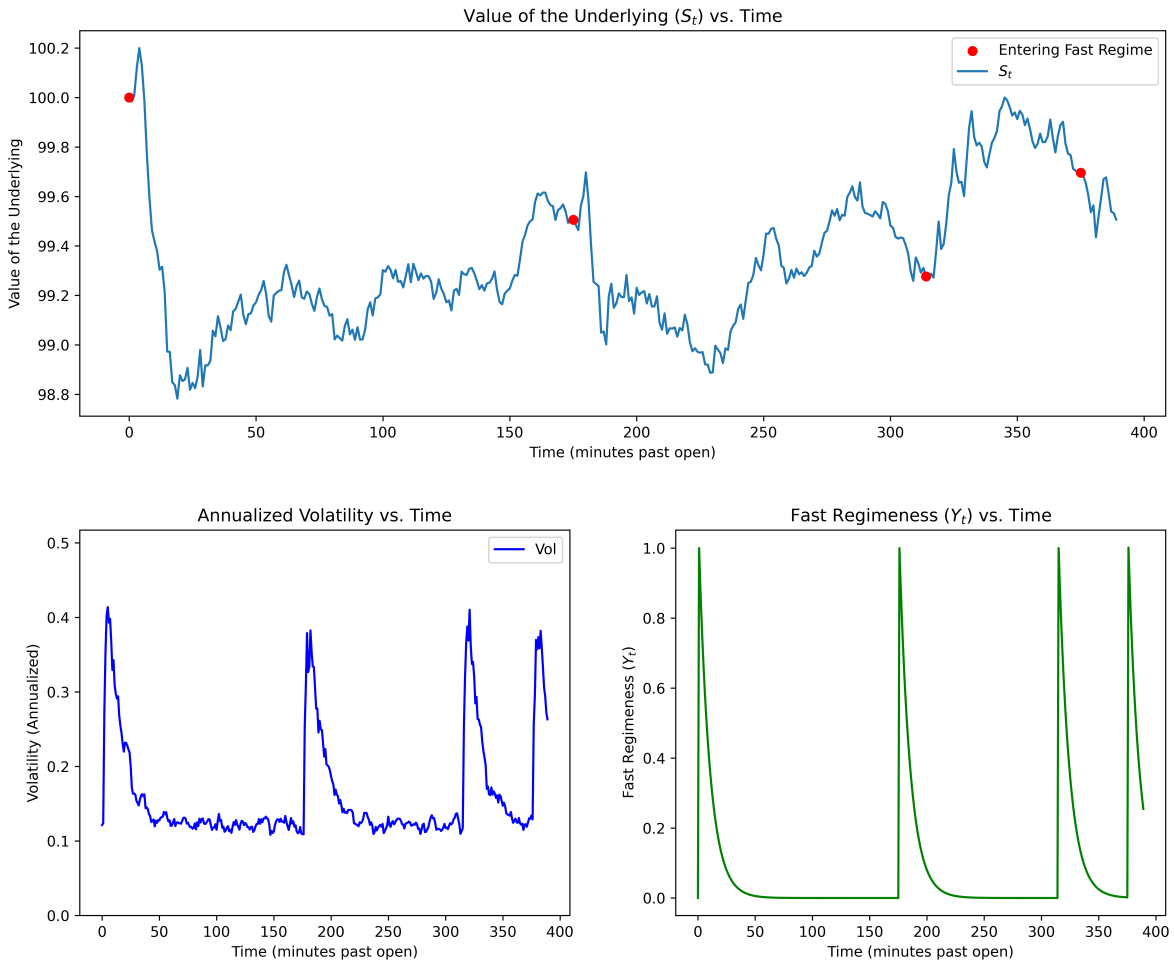

Figure 1. *Sample simulation of the underlying for one day. The top panel is the trajectory of the underlying asset* S_t *with identified entry points into the Fast Regime. The bottom-left panel is the realized annualized volatility during the trading session. The bottom-right is the evolution of the state variable* Y_t *representing Fast Regimeness.*

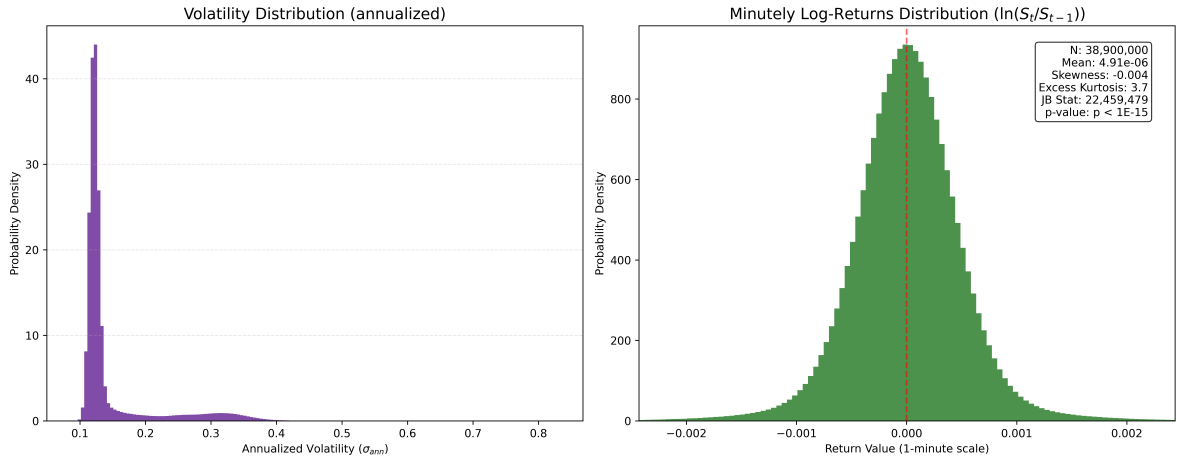

Figure 2. *Aggregate simulation results ($N = 10^5$). The left panel shows the distribution of annualized volatility, while the right panel displays the distribution of minutely log-returns. The high Jarque-Bera statistic confirms that the regime-switching jump process produces heavy-tailed returns for the 1-minute timescale.*

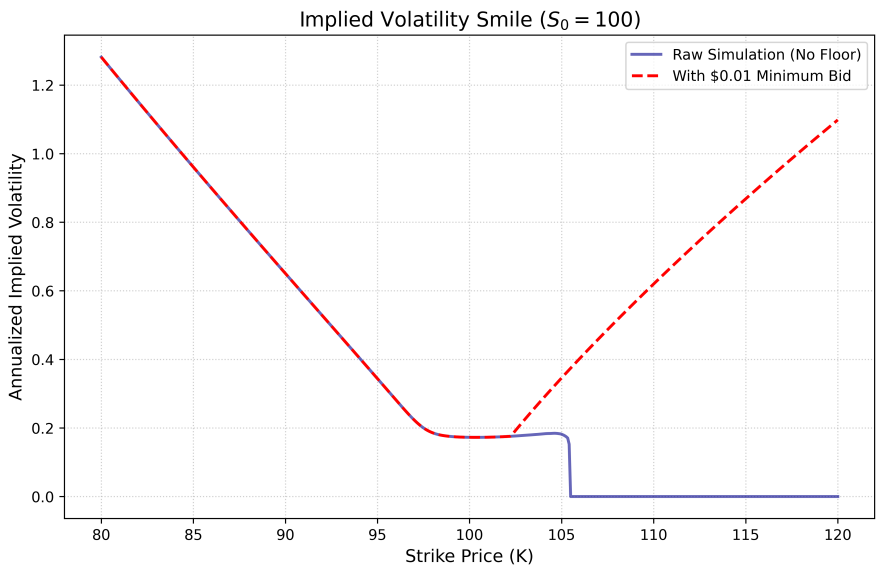

Figure 3. *Annualized implied volatility as a function of strike price (K) ($N = 10^6$) for vanilla call 1-DTE. The simulation incorporates regime-switching volatility, resulting in a pronounced volatility skew. The divergence of the floored curve at extreme out-of-the-money (OTM) strikes demonstrates how a $0.01 minimum bid induces an artificial inflation of implied volatility.*

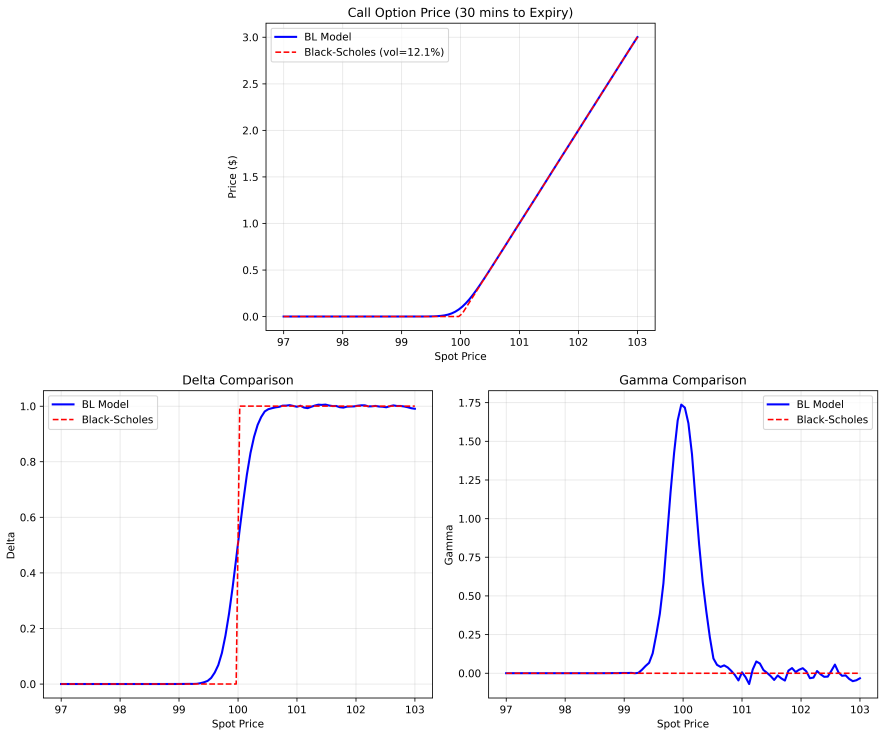

Figure 4. *Option Pricing and Greeks 30 minutes to Expiry* ($N = 10^8$) *for* $K = 100$. *While the the boundary layer model and Black-Scholes models produce nearly identical absolute pricing profiles, the boundary layer model's volatility spikes captures heavier tails. This structural difference is revealed in the Greeks—near expiry, the boundary layer model has well defined deltas and gammas as the volatility blows up in the fast regime. Note, non-smoothness of the delta and gamma are due to Monte Carlo simulation noise. Further, the Black-Scholes model uses the mean volatility from the boundary layer model's slow regime. Last, the ATM Black-Scholes gamma on the bottom right is infinity, not zero.*

Appendix B. Fast Regime Identification Data & Graphs.

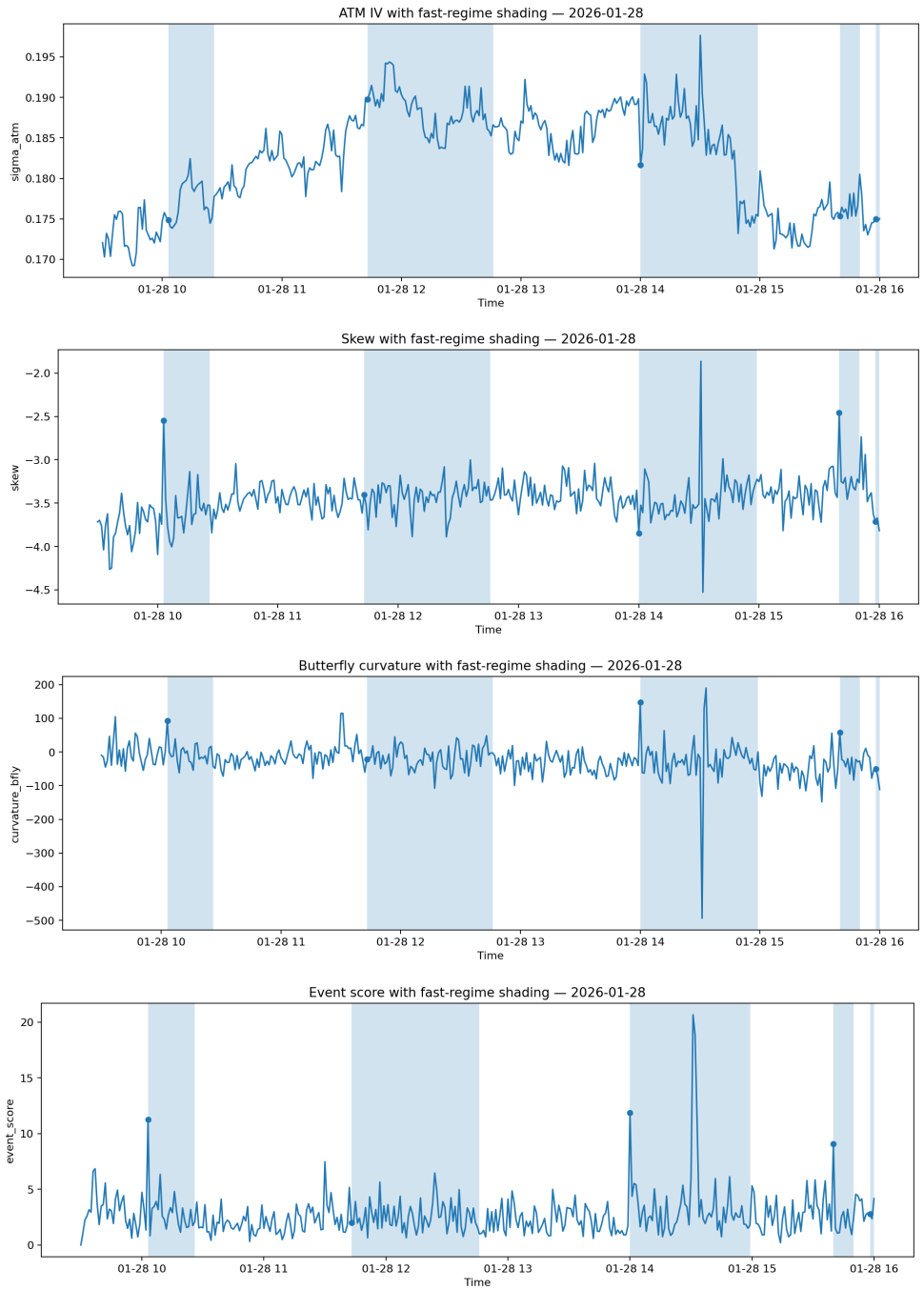

Figure 5. *Intraday smile diagnostics with fast-regime shading for 2026–01–28. From top to bottom: ATM implied volatility* $\sigma_{\mathrm{atm}}(t)$*, local skew, butterfly curvature* $\mathrm{curvature}_{\mathrm{bfly}}(t)$*, and the aggregate event score. Shaded regions indicate minutes classified as fast regime.*

Appendix C. Hedging Performance Data.

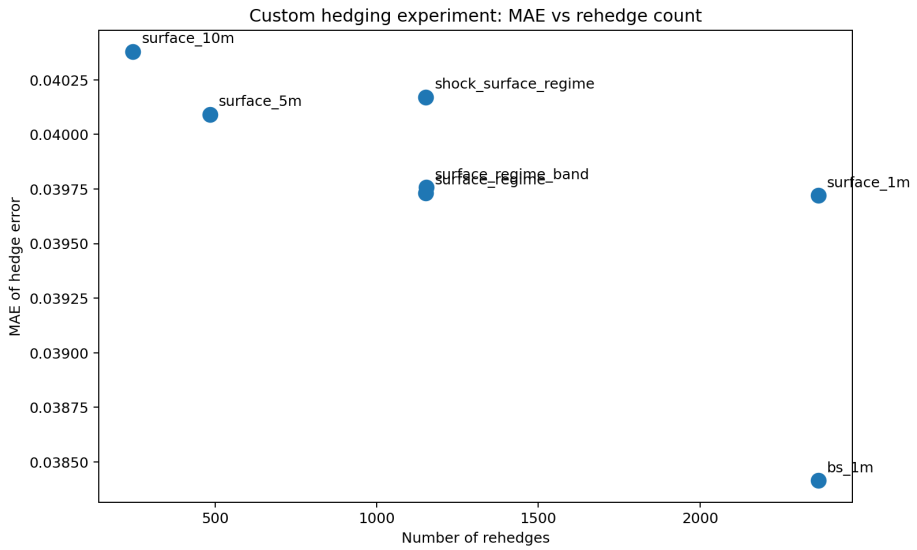

Figure 6. *Efficiency frontier for the custom hedging experiment. The Black–Scholes one-minute hedge attains the smallest raw MAE, while the regime-aware smile hedge nearly matches the fully rehedged smile strategy with roughly half as many hedge updates.*

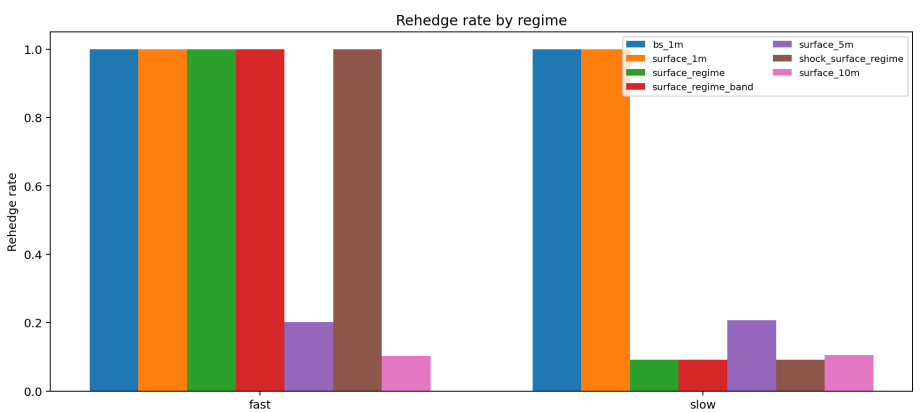

Figure 7. *Rehedge rate by regime. The regime-aware strategies hedge almost every minute in fast periods and sharply reduce hedge activity in slow periods.*